

JS Scope - Practice Questions

1. Function Scope with var

Question:

Declare a variable using `var` inside a function and try to access it outside.

Expected Output:

```
"Inside Function"  
ReferenceError: message is not defined
```

2. Block Scope with let

Question:

Show that `let` variables are block-scoped.

Expected Output:

```
ReferenceError: name is not defined
```

3. Block Scope with const

Question:

Demonstrate `const` inside a block scope.

Expected Output:

```
ReferenceError: pi is not defined
```

4. Global Scope Accessibility

Question:

Show that a globally declared variable is accessible inside functions.

Expected Output:

```
"I am global!"
```

5. Re-declaring var in the Same Scope

Question:

Show that `var` allows redeclaration.

Input:

```
var x = 10;  
var x = 20;
```

Expected Output:

```
20
```

6. Redeclaring let in the Same Scope

Question:

Show that `let` does not allow redeclaration.

Input:

```
let y = 10;  
let y = 20;
```

Expected Output:

```
SyntaxError: Identifier 'y' has already been declared
```

7. Redeclaring const in the Same Scope

Question:

Show that `const` does not allow redeclaration.

Input:

```
const z = 30;  
const z = 40;
```

Expected Output:

```
SyntaxError: Identifier 'z' has already been declared
```

8. Hoisting with var

Question:

Show that `var` variables are hoisted but not initialized.

Expected Output:

```
undefined
```

9. Hoisting with let

Question:

Show that `let` variables are hoisted but not accessible before declaration.

Expected Output:

```
ReferenceError: Cannot access 'b' before initialization
```

10. Hoisting with const

Question:

Show that `const` variables are hoisted but not accessible before declaration.

Expected Output:

```
ReferenceError: Cannot access 'c' before initialization
```

JS Scope - Assignment Questions

1. Using let inside a Loop

Question:

Demonstrate how `let` maintains block scope inside a loop.

Expected Output:

```
0
1
2
ReferenceError: i is not defined
```

2. Using var inside a Loop

Question:

Show that `var` does not maintain block scope inside a loop.

Expected Output:

```
0
1
2
3
```

3. Function Scope Example

Question:

Write a function `functionScopeExample` that demonstrates function scope using `var`.

Expected Output:

```
"Hidden Message"
ReferenceError: secret is not defined
```

4. Function Inside a Block with let

Question:

Show that `let` cannot be accessed outside a block inside a function.

Expected Output:

```
ReferenceError: test is not defined
```

5. Function Using const

Question:

Create a function `constExample` that declares a constant inside a function and tries modifying it.

Expected Output:

```
TypeError: Assignment to constant variable.
```